Task 1: W.S. Gosset & Guinness

See R basic source file (Appendix) for computation details. You are not required to use R. You should note the tools that you use; include source code either as an appendix or separate file. Source code is not graded in this course; if your results are wrong, I may be able to identify the cause.

IPC 1.37 **2 points**

Units…These are percentages, so we should say that…

(a) Mean 122.92
(b) Median 102.5
(c) Not required…but if you ignore this, ask yourself: What value did I obtain from the exercise?

IPC 1.38 **3 points**

(a) Standard deviation 105.73
(b) Quartiles 67.5, 123.5
(c) …

<u>Regarding the quartiles</u>

For question 1, I calculated the quartiles using Minitab and then manually, but got two slightly different answers. Q1 results were 66.5 and 67 but for Q3 Minitab gave me 134.5 and manually I got 129. Is that possible?

Yes, it is possible. The upper quartile, Q_3, is any value that separates the data into two groups: the smallest 75% and the largest 25%. For any set of discrete data (and no matter how large your dataset, it is always a collection of discrete values), there will be a gap between every adjacent (unique) value…so a quartile (or any other percentile) will be somewhat indeterminate –any value in the gap will meet the definition. (If the data has repeated values, then a unique quantile may exist.)

In this example there are 12 distinct data values, so Q_3 is "the" value which separates the smallest 9 values from the largest 3. Here is the ordered data:

$$24\ \ 46\ \ 66\ \ 68\ \ 78\ \ 94\ 111\ 112\ 118\ 140\ 190\ 428$$

Q_3 can be any value between 118 and 140. There are some common conventions: the "obvious" one is to use the mid-point, which is (118+140)/2 = 129. Another convention is to weight the values. If you are interested, I encourage you to research the details – they have to do with continuity corrections, which increase the accuracy of the approximation of continuous distributions such as the normal distribution to discrete data distributions and only have any material impact in small samples such as the one we have here.

$(3 * 118 + 140)/4 \ = \ 123.5$ This is the R result

$(118 + 3 * 140)/4 \ = \ 134.5$ This is the Minitab result

When you see such apparent discrepancies, ask yourself: Do the differences matter? If they do (that will depend on what you are using the data for), then more data and more thinking is needed.

Another contribution:
*I had the same question then I noticed even within R alone, the ?help documentation and threads on stackoverflow suggest there are multiple functions that also may give slightly different results for Q1 and Q3, even varying for odd and even numbers of observations, due to the exact manner in which they are calculated. The box plot for example had to be forced to show outliers using the IQR*1.5 criteria because the IQR differed from the expected calculations.*

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summary(), fivenum(), quantiles(), boxplots.stats
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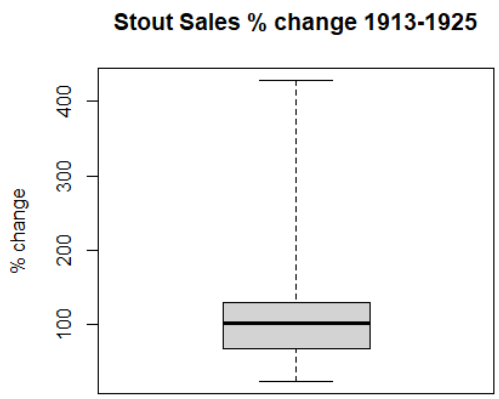
IPC 1.39 **1+3+3+3+1+4 = 15 points**

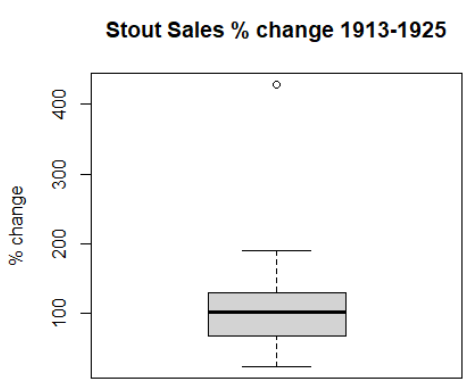
(a) Interquartile range 123.5 – 67.5 = 56
(b) Outliers using the 1.5*IQR rule (p33) $< Q1 - 1.5 * IQR \ or \ > Q3 + 1.5 * IQR$
No low outliers; 1 high outlier: 428 – London
(c) boxplot



Commentary should always be done. Here, we see long right (upper) tail so there is a strong skew. The central portion of the data (between the quartiles) looks approximately symmetric.

A good question to ask yourself with respect to all assignments is: What is the point? There should always be something…

(d) modified boxplot – this is the R standard; software varies in the default settings so this is a good place to explore the software of your choice.



Here we see the presence of a (high value) outlier. Excluding the outlier, the main body of the data distribution looks nearly symmetric. Would you say there is some skew? Are there enough data points to be confident about such a judgement? Remarking on the min, max, median, and quartiles is a good idea too if it is not done elsewhere. The original style boxplot, part (c), does not identify outliers, so the commentary there could only describe the apparent right skew.

(e) …

Stemplot
The decimal point is 2 digit(s) to the right of the |
0 | 257789
1 | 11249
2 |
3 |
4 | 3

Notice the scaling in this stemplot. It gives a sensible (informative) plot which shows the distribution of the data. If your software default gives an uninformative display, explore alternative settings. Remember that the goal is to extract information from data, not just "run a function".

(f) Similar to what we have remarked previously, now particularized to the Guinness data. The outlier is easy to spot in both displays; I find the symmetry of the stout data distribution easier to assess in the boxplot than the stemplot.

Note: The old style boxplot (in (c)) is generally not used now in scientific work (but it is often seen in less technical publications) because the modified form showing outliers (if there are any) is so much more useful.